MIS 201 · WEEK 6 · WEDNESDAY 10/28

Lab 6: You Are the Auditor

Recap, then the lab, then the record and the debate. Pairs, one machine.

Today

2:00	**Recap of Monday**	4 min
2:04	**Lab 6, sections 1 to 4**	52 min
2:56	Break	5 min
3:01	**The record: before and after the algorithms**	8 min
3:09	**Debate: prep and rounds (follow the Gilman Rules!)**	37 min
3:46	**Wrap-up and attendance**	4 min

DUE FRIDAY 10/30, 11:59 P.M.

- Lab 6 writeup, five items
- Quiz on Chapter 6
- Project team: four names and one paragraph

Section 5 and the writeup are for after class. Jot writeup notes as you go today.

Recap from Monday

1. **The classifier**
   Learns who got the job, not who deserved it. A threshold turns a score into yes or no.

2. **The objectivity trap**
   Consistency isn't neutrality. The bias moved into the data.

3. **Proxies**
   Zip code for race, graduation year for age. Deleting the field doesn't delete the pattern.

4. **Two kinds of wrong**
   The company sees false positives. Nobody counts the invisible pile.

5. **Who answers**
   iTutorGroup settled. Mobley v. Workday is still in court. Hiring has no right to reasons.

6. **The audit**
   A matched pair finds what it reads. Four-fifths finds whether the outcome skews.

WHAT DO YOU THINK?

Tell the person next to you, in one sentence, why deleting the gender field didn't fix Amazon's model. Go.

Lab 6: today you train three classifiers, watch two learn old bias, and build a résumé scanner with Claude. Then the room debates: should a model or a loan officer make the call?

Part 1: the lab

52 MIN · PAIRS

1	**Teach it to sort: a spam filter**	10 min	done by 2:14
2	**Same machine, car loans: the loan desk**	12 min	done by 2:26
3	**Same machine, résumés: the résumé scanner**	15 min	done by 2:41
4	**Build your own résumé scanner with Claude**	15 min	done by 2:56
5	**Under the hood**	after class	at home

HOW IT WORKS

- Pairs, one machine.
- Sections in order; each tool tells you the next step.
- Jot writeup notes as you go.
- Section 4: delete your phone, email, and address before scanning your own résumé.
- Rather not use AI? Section 4 has a by-hand keyword audit.
- Log out of Claude when you're done.

Section 1 is 2.1's classifier. Sections 2 and 3 are 2.2 and 2.3. Your audit tools are 2.6's.

Sections 1 to 3: what to watch for

LAB · SECTIONS 1 TO 3

1 · MINI-SPAM

The cutoff is your decision.

Find a cutoff that loses no real email, then one that lets no spam through. Is there one that does both? Look at the ties. Then flip one old email's label: what did the filter believe?

2 · THE LOAN DESK

Same file, 26 points apart.

Riley and Morgan match on income and credit. Make your call, then trace the gap back to the loan officers. Write the four-fifths ratio at all three steps. Why didn't deleting neighborhood fix it?

3 · THE RÉSUMÉ SCANNER

The pair that comes back clean.

Watch the keyword gate, then the word "women's." Run the graduation-year pair before and after you delete it. When the pair matches, is the scanner fair? Check the ratio before you answer.

Each tool fills in a record table. The writeup asks for those numbers, so check they're there before you move on.

Section 4: build your own résumé scanner

LAB · SECTION 4

1. Copy the build prompt into a new chat at claude.ai. Send it once.
2. Try it on Jordan: paste the posting and résumé from the page. Did it figure out “database queries” means SQL?
3. Scan yourself: a real posting plus your résumé, contact details deleted. Out of time? Finish this at home.
4. Keep it: save the page or its link for writeup items 3 and 5.

YOU’RE THE APPLICANT

Claude writes the code.

You decide which suggestions are true. Never claim a skill you can’t back up.

BEFORE YOU START

Every scan uses your Claude message limit. If the page can’t reach Claude, turn on AI artifacts in settings. No AI? Do the keyword audit by hand.

Missing keywords you already have are easy wins. The rest is a list of things to learn.

Section 5: where the bias hid

AFTER CLASS

The code was the same every time you switched. Every toy model today was a weight per field, a score, and a cutoff. Claude's scanner has no weights you can read, which is why you test it from the outside.

IN THE LABEL

An assumption nobody wrote down.

"Approved" was assumed to mean creditworthy; "called back," a good candidate. Someone chose that column.

IN THE PROXIES

Fields doing other fields' work.

Branch carried neighborhood. Years of experience carried age, after the fields themselves were gone.

IN WHAT NOBODY MEASURES

The pile no one follows up.

A real bank never learns whether the people it turned down would have repaid, so its model never learns from those mistakes.

Underneath all three: the assumption that the past is a fair guide to the future.

BREAK · 5 MINUTES

Back in five

When you come back: the record before and after the algorithms, this week's case, the Gilman Street rules, then your side. Left half of the room is Side A, right half is Side B.

Before any algorithm: the human record

RECORD · 1 OF 3

1930s — **Redlining**
Neighborhoods graded largely by who lived there. 74% of "hazardous" areas are still low-to-moderate income.

1947 — **The GI Bill**
Of 3,229 VA-guaranteed loans in 13 Mississippi cities, 2 went to Black veterans.

Before 1974 — **Women and credit**
Lenders could demand a male co-signer and ignore a wife's income on a mortgage.

1992 — **The Boston Fed**
With similar finances, Black and Hispanic applicants were about 60% more likely to be denied.

2004 — **The résumé study**
White-sounding names got 50% more callbacks on identical résumés.

2011 — **Countrywide**
Officers' and brokers' "judgment" cost 200,000+ Black and Hispanic borrowers more. $335M.

2021 — **108 big employers**
83,000 fake applications: distinctively Black names got about 8% fewer contacts.

2023 — **City National Bank**
The Justice Department's largest redlining settlement, $31M+. The bank disagreed.

Notice the pattern: people with discretion. Side A leans on this. Side B says it's exactly what models learn from.

When the algorithms did better

RECORD · 2 OF 3

			THE CATCH
2009–15	**Algorithmic lenders**	27–37% smaller racial gaps in FHA mortgage pricing than face-to-face lenders.	*Gone by 2018–19.*
2018	**Managers vs. the test**	Managers who overruled a hiring test made worse hires.	*Measured tenure, not bias.*
2018	**Judges vs. an algorithm**	Up to 42% fewer people jailed, no rise in crime, smaller racial gaps.	*A simulation, and bail, not loans.*
2020	**A résumé screener**	Picked candidates 14% likelier to get offers, and more women and minorities.	*One company, working paper.*
2025	**Hiring as exploration**	More than doubled interviews for Black and Hispanic applicants, 10% to 23%.	*A standard screener cut that under 5%.*

Side A leans on this. Every time an algorithm beat people, someone designed it carefully and checked it against real outcomes.

When the algorithms made it worse

RECORD · 3 OF 3

2015	**Amazon's scanner**	Learned to penalize "women's" and two all-women's colleges. Scrapped.
2016	**COMPAS**	Black defendants who didn't reoffend were flagged high risk 45% of the time; white, 23%.
2018	**The coded gaze**	Face analysis misread darker-skinned women up to 34.7%; lighter-skinned men, 0.8%.
2018–19	**The edge disappeared**	Algorithmic lenders' racial gaps caught up with everyone else's.
2020	**Howard vs. NYU**	Same borrower, different college: about $3,499 more over a five-year loan.
2022	**Smarter, not fairer**	Better credit models; Black and Hispanic borrowers gained less.
2024	**3 million matched pairs**	AI models preferred white-associated names 85% of the time.

Side B leans on this. None was told to discriminate. Nobody outside would have known without someone running a test.

This week's case: Amazon's résumé scanner

CASE OF THE WEEK

10 years

of résumés, mostly from men, taught the scanner what a good hire looked like.

It learned to penalize the word "women's" and to downgrade graduates of two all-women's colleges. Engineers edited out those terms, couldn't be sure it wouldn't find other ways to discriminate, and scrapped it.

SIDE A USES IT

The bias came from Amazon's human hiring. The model made it visible, and engineers could look inside and find it.

SIDE B USES IT

The model took that bias and scaled it to every applicant, and its own builders couldn't fix it.

Source: Jeffrey Dastin, Reuters, October 10, 2018.

Debate rules: the Gilman Street rules

APPLY THESE TODAY

Same rules as always, borrowed from 924 Gilman Street. Here's what each one means in today's debate.

NO RACISM · NO SEXISM · NO HOMOPHOBIA · NO TRANSPHOBIA	Argue with ideas, never at people. Extra care today: the topic is discrimination itself.
NO FIGHTING	No attacking a position without evidence. "That's wrong" isn't a point. "That's wrong, because the Boston Fed found…" is.
NO STAGEDIVING	No interrupting, piling on, or ambushing. One speaker at a time; let them finish.
NO MAJOR LABEL BANDS	No blind love or hate for a company or a profession. Amazon, banks, loan officers, AI: give reasons, not fandom.

The debate: model or loan officer?

37 MIN

The motion: **A bank's audited model, not a loan officer, should make the final call on every car loan and mortgage.**

SIDE A · LEFT HALF OF THE ROOM

For: the model makes the call

The bank's model decides every application. An outside auditor tests it every year, in public, and every no comes with its main reasons.

What you have to prove: Prove a model can be caught and fixed faster than a person, even though Amazon couldn't fix theirs.

SIDE B · RIGHT HALF OF THE ROOM

Against: a person makes the call

A trained loan officer makes the final decision. A model can give a score, but a person signs, and writes down why when they go against it.

What you have to prove: Prove your loan officers would be fairer than the people in the human record.

Resolve it: Amazon's scanner learned its bias from Amazon's own hiring. Is a model fairer than the people it learned from, or the same bias running faster?

Side A talking points: the model decides

SIDE A

1	**“People were biased first, and they still are.”**	GI Bill: 2 of 3,229 loans. Boston Fed: about 60% more denials. Countrywide: $335M.
2	**“Algorithms were fairer than face-to-face lenders.”**	27–37% smaller racial gaps in FHA mortgage pricing, 2009–15.
3	**“When people overrule the tool, they usually get it wrong.”**	Managers who overrode a hiring test made worse hires.
4	**“A model’s bias can be found. A person’s mostly can’t.”**	Named in Gender Shades, all three companies cut their gaps within 7 months. Human bias took 83,000 fake applications to measure.
5	**“Built carefully, an algorithm opens the door wider.”**	Interviews for Black and Hispanic applicants, 10% to 23%. Bail: up to 42% fewer jailed.

Count off 1 to 5 on your side. That’s your point. The comeback to expect is on the lab page.

Side B talking points: a person decides

SIDE B

1	**“A model copies its teachers, at scale.”**	Amazon’s scanner learned to penalize “women’s” from ten years of mostly men’s résumés.
2	**“You can’t just delete the bias.”**	Amazon edited out the words, couldn’t guarantee there weren’t others, and scrapped it.
3	**“Smarter isn’t the same as fairer.”**	Better credit models widened rate gaps. COMPAS flagged Black non-reoffenders high risk 45% vs. 23%.
4	**“AI still sees some people worse than others.”**	AI screeners preferred white-associated names 85% of the time. Face analysis: 34.7% vs. 0.8% errors.
5	**“When someone tested a real AI lender, it failed.”**	Same borrower, Howard instead of NYU: about $3,499 more over a five-year loan.

Count off 1 to 5 on your side. That’s your point. The comeback to expect is on the lab page.

How the debate runs: prep

12 MIN

1. **Get your point** 1 min

   Groups of 2 or 3. Count off 1 to 5 on your side.

2. **Check the evidence** 3 min

   Click your point's tag on the lab page, reread its timeline entries, open one source.

3. **Pressure-test it with Claude** 5 min

   Copy the prompt on the page, fill in the brackets, send. Check any study it cites. Close it, log out.

4. **Write your notes** 3 min

   In your own words, on paper or in a doc.

YOUR NOTES

1. Your point, in one sentence
2. The evidence and its source
3. The comeback you expect
4. Your answer to it

THE RULE ON AI

Use it to research. Don't read from it.

How the debate runs: the rounds

25 MIN

Team up	3 min	Join the other groups on your side. Pick your best two points, an opener, and a rebutter.
Openings	6 min	Side A, then Side B. About 45 seconds each: your claim, your evidence, and its source.
Huddle	2 min	Rebutters pick one point the other side made.
Rebuttals	6 min	Side B first. Answer one specific point, with evidence. No new arguments.
Switch	4 min	What was the other side's best point? Someone new argues it in one sentence.
What would change your mind?	4 min	Out loud. Name the specific evidence you'd have to see.

THE GILMAN RULES, APPLIED

- One speaker at a time. Let them finish.
- Every point needs evidence and a source.
- Answer the argument, not the person.
- Reasons, not fandom.

Whoever names real evidence that would move them wins, whichever side they were handed.

BEFORE YOU LEAVE

Upload your notes

Attendance = your notes from today, uploaded to Canvas right after class.

At home: section 5 (a short read), your own résumé scan if you didn't finish, and the writeup. Item 2 is your loan desk and résumé scanner numbers; item 3 is the scanner you built; item 4 is the debate, with at least one source; item 5 is your commitment card from your own scan.

Friday 10/30, 11:59 p.m.: Lab 6 writeup, Chapter 6 quiz, and your project team, all on Canvas.

Log out of Claude before you go. Shared machines.